

Features

ASUS and Palit were the two ATI vendors who sent us their HD4870 offerings. Both were overclocked over the 750 MHz that ATI recommends as a stock speed. The Palit HD 4870 SONICs core is clocked at 775 MHz while the memory runs at the recommended speed i.e. 3,600 MHz. Palit uses a custom cooler – it features two heatpipes, and two fans – we feel this cooler is much more efficient than ATI's stock cooler, and looks good too. This card ran cooler than the ASUS EAH 4870 TOP, which was overclocked to a whopping 825 MHz. Palit's bundle was sparse with the usual power connectors and an S-Video cable. ASUS includes the usual goodies in the form of a leather mousepad and CD pouch, but no games. Both vendors include an HDMI adapter.

Performance

Shockingly both 4870s hold their own against the GTX 280s in *Crysis* – a very, very good result for ATI – since this game engine is very new, and is likely to be indicative of future performance for DX10 games. *UT 3* proves to be CPU limited on these cards, although they cannot match the GTX 2xx family. Both these cards also perform well in *Company Of Heroes*, and manage to out-perform even the HD4870 x2. However the GTX 280 and GTX 260 are still too tough a proposition to tackle. *World In Conflict* and *F.E.A.R* are easy for these cards although they are slower than NVIDIA's offerings in the high-end segment. But the most shocking factor for us was the price – these cards are nearly 50

per cent cheaper than the GTX 280s and close to 15 per cent cheaper than the GTX 260 cards. When you consider this the HD 4870 makes a very good buy for someone looking at a very fast card for under the Rs 20,000 mark. Palit's HD 4870 SONIC is just superbly priced at an MRP of Rs 17,900 – you might get it even cheaper in the market.

ATI Radeon HD 3870 x2

No V8 this

One of the biggest mistakes a competitor can make is to react to a competing product with something that is half baked. ATI needed something to counter the G80 and all its variants which not only ate into its sales but also established itself as the benchmark for DX10 graphics cards. The Radeon HD3870 was touted as an 8800 GTX killer, but even 320 SPs seemed unable to counter the solid across-the-board performance that NVIDIA was getting with its cards. The HD3870 x2 was born out of desperation and as any birthing under such circumstances it was a stop gap at best, a drastic failure at worst. This was however, ATI's first multi-GPU card since awhile, and it paved the way for the mighty HD 4870 x2. With 640 SPs the 3870 x2 seemed to be unbeatable on paper. It also sported GDDR4 memory, which was the second time ATI went with this largely discarded memory standard – the X1950XTX being the only other GDDR4 sporting graphics card around. NVIDIA hasn't touched GDDR4 thus far, and isn't likely to either. The HD3870 x2 had 1 GB of GDDR4

clocked at 2.0 GHz, and it was a CrossFire solution that didn't need a CrossFire motherboard – it worked just fine with a single PCIe slot.

Features

ASUS sent us a very fancy looking version of the 3870 x2 called the EAH 3870 x2. Basically, this card had four DVI ports instead of the normal two, and sported a very different looking cooler which is much lighter than the stock ATI cooler, and has two fans instead of one. The other card was from Sapphire. This card was based around a stock ATI cooling design. Both cards came clocked at 825 MHz for each core – this also ensures these cards stay toasty throughout our tests, and the Sapphire HD 3870 x2 also ran hot throughout the tests. In fact this card heated within five minutes of booting into Windows Vista. The ASUS card was cooler by an aggregate of 15 degrees, which suggests that ASUS' cooler is really a godsent for users. ASUS provided the usual accessories – a leather CD pouch and a mousepad, also bundled was an HDMI adapter, and Component and S-Video connectors. *Company Of Heroes* is a older game, but we'll give ASUS a plus for bundling one of our all time favourites. The Sapphire card didn't have any games, although an HDMI adapter and the usual S-Video cables were bundled.

Performance

In terms of fps the 3870 x2 isn't too hot. It does manage to outpace the newer HD 4870 cards but only in one game - *S.T.A.L.K.E.R.*, and this isn't too

indicative of performance. The fact that a single GPU HD 4870 is able to thrash it by a good margin in all of the games including taxing ones like *Crysis*, *Company Of Heroes* and *World In Conflict*, not to mention 3D Mark Vantage, showcases the shortcomings of the Radeon 3870 architecture. It's equally incomparable to the GTX 2xx series, which simply massacre it. Stay away from both these cards – they are simply obsolete – there's no other word for it.

The Grand Finale

All the GTX 280s are any gamers dream tool to wreak some serious fps (no pun!) damage. The XFX GX-280-ZDD9 was officially the fastest card around – based around the GTX 280 architecture this card is blazingly fast. It gets our coveted *Best Performance* award. But at Rs 45,000 it's not necessarily the best card around to buy. Our *Digit Best Value* offering was the superb Palit HD 4870 SONIC – with superb performance, a great cooler and a price tag of Rs 17,900 this card is your best bet for an under-Rs 20,000, high-end gaming setup. The fact that is close on the heels of the much costlier GTX 280 based cards is in itself a huge plus for it and this in itself is a compelling reason to buy one.

Mid-range Graphics Cards

The mid range has always been more exciting for most of us than the previous category for one reason. It's much better to see and use cards that you can actually afford than playing with costly stuff you know you can never buy. Most of our readers would also be more interested in this category. These cards are more general purpose; after all nobody would buy a Rs 30,000 plus costing, 240 SP, GTX 280-based card for an HTPC. But a mid-range card priced around the Rs 10,000 mark is a sweet deal – you can play all your games at medium resolutions with high levels of detail, plus these cards are very suitable for powerful HTPCs, not to mention they consume much less power and generate less heat.

We were surprised to see the prices of the NVIDIA 9800GTX crash to the extent of bringing it falling head-long into this category – if this test were conducted in May these cards would be in the high-end category. The most interesting entrants in this category are the very new NVIDIA GeForce 9800GT cards, which are 65nm replacements of NVIDIA's best selling GeForce 8800GT cards. Equally appealing is the new ATI Radeon HD4850, which has come from nowhere and taken top position in terms of performance. The mouth watering prices of these cards have to be seen to be believed.

NVIDIA GeForce 9800 GTX / 9800 GT / 8800 GT

The new mid range

The 9800 GTX isn't a new GPU – it's been around for a few months. Touted as the replacement for the mighty 8800 GTX, this GPU wasn't close to the G80 in terms of architecture, as NVIDIA had applied a lot of cost cutting measures. It claimed the 384-bit memory bus on the 8800 GTX was doing nothing for

| Mid Range GPUs                 | NVIDIA 9800 GTX+ | NVIDIA 9800 GTX | NVIDIA 9800 GT | NVIDIA 8800 GT | ATI HD 4850 | ATI 3870 |
|--------------------------------|------------------|-----------------|----------------|----------------|-------------|----------|
| Stream Processors              | 128              | 128             | 112            | 112            | 800         | 320      |
| Texture Addressing / Filtering | 64 / 64          | 64 / 64         | 56 / 56        | 56 / 56        | 40          | 16       |
| ROPs                           | 16               | 16              | 16             | 16             | 16          | 16       |
| Core Clock Speed (MHz)         | 738              | 675             | 600            | 600            | 625         | 775      |
| Memory Bus Width (bits)        | 256              | 256             | 256            | 256            | 256         | 256      |
| Memory Frequency (MHz)         | 2200             | 2200            | 2000           | 1800           | 1986        | 2250     |
| Memory Type                    | GDDR3            | GDDR3           | GDDR3          | GDDR3          | GDDR3       | GDDR4    |
| Amount of memory (MB)          | 512              | 512             | 512            | 512            | 512         | 512      |

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